



Curriculum Vitae

• Personal information

Name Dr. Ricardo Martínez-García.
Date of birth 02/02/1988.
Affiliation Center Advanced Systems Understanding – Helmholtz-Zentrum Dresden-Rossendorf.
Address Conrad-Schiedt-Str. 20, D-02826 Görlitz.
Website <https://sites.google.com/view/rmtzgarcia>
Google Scholar <https://scholar.google.com/citations?user=tak0qSUAAAAJ&hl=en>

• Education

PhD 2014 Physics. University of the Balearic Islands at the Institute for Cross-Disciplinary Physics and Complex Systems (IFISC), Spain. June 2014.
- Including a three-month research stay at Smithsonian Conservation Biology Institute (USA).
MSc 2011 Physics. University of the Balearic Islands, Spain. September 2011.
BSc 2010 Physics. University of La Laguna, Spain. Specialization in Theoretical Physics.

• Appointments

2022- Young Investigator Group Leader. Center for Advanced Systems Understanding (CASUS) and Helmholtz-Zentrum Dresden-Rossendorf. Görlitz, Germany
2022- International faculty at Princeton / CUNY Center for the Physics of Biological Function, Princeton USA.
2020-22 SIMONS Associate at the Quantitative Life Sciences section at Abdus Salam ICTP. Trieste, Italy.
2019-22 SIMONS-FAPESP Assistant Professor. International Center for Theoretical Physics – South American Institute for Fundamental Research (ICTP-SAIFR) & Institute for Theoretical Physics, State University of São Paulo. São Paulo, Brazil (tenured granted in March 2022).
2016-19 Life Sciences Research Foundation Postdoctoral Fellow. Department of Ecology and Evolutionary Biology, Princeton University, USA.
2014-16 Postdoctoral Research Associate. Department of Ecology and Evolutionary Biology, Princeton University, USA.
2010-14 CSIC JAE-Predoc Fellow. Institute for Cross Disciplinary Physics and Complex Systems (IFISC), Palma de Mallorca, Spain.

• Awards & Honors

2023 Junior Scientific Award of the Complex Systems Society.
2021 Affiliated member of the Brazilian Academy of Sciences (Affiliated members are young scientists < 40 years old elected by the regional presidencies of the Academy for five-year memberships).
2015 Honoric Mention of the Spanish Council of Research (CSIC) for research achievements during the academic year 2014/2015.
2014 Montserrat Casas award in the category of Sciences, Engineering, and Architecture. Best PhD research in the Balearic Islands, Spain.
2013 Banco Santander-UIB award. Best PhD students in the University of the Balearic Islands.
2006 Bronze Medal, Spanish Physics Olympiad.

• Competitive funding

As principal investigator

- 2025-28** *Inferring desertification risk from vegetation patterns using functional network analysis*. CASUS Open Project (~**5000 EUR and one PhD student position**). Co-PI: Prof. Dr. Viola Priesemann.
- 2022-27** *Self-organization and collective phenomena in complex biological systems*. CASUS Young Investigator Group. Funded by CASUS-HZDR, budget line FWU-C (~**1.5M EUR, including two postdoc positions, own salary, and 125.000 EUR for research costs**).
- 2019-23** *Towards a mechanistic theory for biological pattern formation and their ecosystem-management opportunities*. São Paulo Research Foundation, FAPESP 2019/24433-0. (~**1.1M BRL / ~170.000 EUR, including 1 PhD and 2 Master Fellowships**).
- 2020-22** *The role of environmental flows on the dynamics of microbial communities*. Simons Foundation Grant number 284558FY19 (**23.000 EUR**).
- 2020-22** *Understanding the ecology and evolution of microbial social behaviors in turbulent environments: cooperation as a case study*. Instituto Serrapilheira, Serra-1911-31200 (**100.000 BRL / ~15.000 EUR**).
- 2016-19** *Using Dictyostelium discoideum to understand the role of ecology and multiple fitness tradeoffs in the evolution of multicellularity*. Gordon Betty Moore Foundation, GBMF2550.06 (**180.000 USD**).

Academic host for competitive fellowship holders

- 2026-27** Guilherme S. Costa. International Postdoctoral Fellowship. Brazilian National Council for Scientific and Technological Development (CNPq), Grant #401209/2024-5 (**168.159,72 BRL / ~26.000 EUR**).
- 2025-26** Ian Magalhães. International Postdoctoral Fellowship. Brazilian National Council for Scientific and Technological Development (CNPq), Grant #401209/2024-5 (**168.159,72 BRL / ~26.000 EUR**).
- 2024** Matheus Bongestab. DAAD Co-financed short-term research grant Brazil, Ref no. 91900240. (**5.750 EUR**).
- 2023-24** Rafael Menezes. International PhD stays. Brazilian Ministry of Education CAPES (Coordination for the Improvement of Higher Education Personnel)-PRINT, Fellowship #88887.889773/2023-00. (**19.310 EUR**).
- 2023** Nathan Silvano. International PhD stays. Brazilian Ministry of Education CAPES (Coordination for the Improvement of Higher Education Personnel)-PRINT, Fellowship #88887.803415/2023-00. (**16.150 EUR**).

As an associated researcher.

- 2022-27** *ICTP South American Institute for Fundamental Research: a regional center for Theoretical Physics*. Role: responsible for leading the research line on complex systems and biological physics with access to one postdoc hiring and travel funds. São Paulo Research Foundation, FAPESP, 2021/14335-0 (**11.325.440 BRL and 532.000 USD; total ~ 2.2M EUR**). PI: Prof. Dr. Nathan Berkovits.
- 2013-14** *Complex Systems Physics: Information, Technology, Society, and Ecology (INTENSE@COSYP)*. Role: contributed to the working package related to ecological systems. Spanish Ministry of Economy and Competitiveness, FIS2012-30634 (**515.460 EUR**). PI: Prof. Dr. Maxi San Miguel.
- 2011-14** *Física Interdisciplinar de Sistemas Complejos (FISICOS)*. Role: contributed to the working package related to ecological systems. Spanish Ministry of Education and Science, FIS2007-60327 (CONSOLIDER C). (**1.318.900 EUR**) Principal Investigator: Maxi San Miguel.

• Publications

An updated bibliometric summary is available at my [Google Scholar](#).

Total: 40 peer-reviewed articles, 3 book chapters, and 7 preprints under review. 18 first or co-first author, 14 as senior author. Last or (co)-first author in approximately 65% of the publications.

Underlined names indicate mentee co-authors; * indicates shared first authorship (most often involving

Published

- 40 **M. Bongestab, D. Pinto-Ramos, & R. Martínez-García[♦]**. Self-organized vegetation patterns promote persistence of plant–pollinator mutualisms under environmental stress. *PLOS Ecosystems*, *In press*. Preprint available at: [10.48550/arXiv.2512.00254](https://doi.org/10.48550/arXiv.2512.00254).
- 39 A. Martínez, *et al.*, *Flagship species as subterranean monitoring tools: a case study in anchialine ecosystems in Lanzarote (Canary Islands)*. *Biodiversity and Conservation*, 35(8), 211 (2026). [10.1007/s10531-026-03389-7](https://doi.org/10.1007/s10531-026-03389-7)
- 38 **B. Garcia de Figueiredo, I. Silva, M.J. Noonan, C.H. Fleming, W.F. Fagan, J.M. Calabrese, R. Martínez-García[♦]**. How animal movement influences wildlife-vehicle collision risk: a mathematical framework for range-resident species. *Journal of Theoretical Biology* 627, 112471, (2026). [10.1016/j.jtbi.2026.112471](https://doi.org/10.1016/j.jtbi.2026.112471)
- 37 **D. Pinto-Ramos, R. Martínez-García[♦]**. How spatial patterns can lead to less resilient ecosystems. *Proc. Natl. Acad. Sci. USA*, 123 (14) e2511994123 (2026) [10.1073/pnas.2511994123](https://doi.org/10.1073/pnas.2511994123)
- 36 J. van der Voort[♦], **R. Martínez-García**, and A. Doelman. *Vegetation pattern formation induced by local growth outpacing susceptibility to non-local competition*. *Journal of Theoretical Biology*, 626, 112441, (2026) [10.1016/j.jtbi.2026.112441](https://doi.org/10.1016/j.jtbi.2026.112441)
- 35 **R. Menezes, J.M. Calabrese, W.F. Fagan & R. Martínez-García[♦]**. The range-resident logistic model: a new framework to formalize the population-dynamics consequences of range residency. *Ecology Letters* 28(12), e70269 (2025). [10.1111/ele.70269](https://doi.org/10.1111/ele.70269)
- 34 **N.O. Silvano, J. Valeriano, E. Hernández-García, C. López, R. Martínez-García[♦]**. Flow spatial structure determines pattern instabilities in nonlocal models of population dynamics. *Communications Physics* 8, 326 (2025). [10.1038/s42005-025-02246-3](https://doi.org/10.1038/s42005-025-02246-3)
- 33 **A. Surendran*, D. Pinto-Ramos*, R. Menezes*, R. Martínez-García[♦]**. Spatial moment dynamics and biomass density equations provide complementary, yet limited, descriptions of pattern formation in individual-based simulations. *Physica D*, 477, 134703, (2025). [10.1016/j.physd.2025.134703](https://doi.org/10.1016/j.physd.2025.134703)
- 32 **W.F. Fagan[♦], A. Krishnan, Q. Liao, C.H. Fleming, D. Liao, C. Lamb, B. Patterson, T. Wheeldon, R. Martínez-García, J. Menezes, M.J. Noonan, E. Gurarie, J.M. Calabrese**. Intraspecific encounters can lead to reduced range overlap. *Movement Ecology* 12, 58, (2024). [10.1186/s40462-024-00501-w](https://doi.org/10.1186/s40462-024-00501-w)
- 31 **V. Dornelas*, P. de Castro*, J.M. Calabrese, W.F. Fagan, R. Martínez-García[♦]**. Movement bias in asymmetric landscapes and its impact on population distribution and critical habitat size. *Proc. Royal Society A* 480, 20240185, (2024). [10.1098/rspa.2024.0185](https://doi.org/10.1098/rspa.2024.0185)
- 30 **C. Cabal*, G.A. Maciel*, R. Martínez-García[♦]**. Plant antagonistic facilitation across environmental gradients: a soil-resource ecosystem engineering model. *New Phytologist* 244, 670-682, (2024). [10.1111/nph.20053](https://doi.org/10.1111/nph.20053)
- 29 **E.H. Colombo, R. Martínez-García, J.M. Calabrese, C. López, E. Hernández-García[♦]**. Pulsed interactions unify reaction–diffusion and spatial nonlocal models for biological pattern formation. *Journal of Statistical Mechanics: Theory and Experiment*, 3, 034001 (2024). [10.1088/1742-5468/ad2b57](https://doi.org/10.1088/1742-5468/ad2b57)
- 28 **D.C.P. Jorge, R. Martínez-García[♦]**. Demographic effects of spatial patterns in the presence of a component Allee effect. *Journal of the Royal Society Interface* 21, 20240042 (2024). [10.1098/rsif.2024.0042](https://doi.org/10.1098/rsif.2024.0042)
- 27 **E. Ser-Giacomi[♦], R. Martínez-García, S. Dutkiewicz, M.J. Follows**, *Dilution-driven heterogeneity enhances plankton productivity in Lagrangian ecosystems*. *Nature Communications*, 14, 6092 (2023). [10.1038/s41467-023-41469-2](https://doi.org/10.1038/s41467-023-41469-2)
- 26 **R. Martínez-García[♦], C. Cabal, J.M. Calabrese, E. Hernández-García, C.E. Tarnita, C. López, J.A. Bonachela**. Integrating theory and experiments to link local mechanisms and ecosystem-level consequences of vegetation patterns in drylands, *Chaos, Solitons & Fractals*, 166, 112881 (2023). [10.1016/j.chaos.2022.112881](https://doi.org/10.1016/j.chaos.2022.112881)
- 25 **M.P. Veldhuis[♦], R. Martínez-García, V. Deblauwe, V. Dakos**. Remotely-sensed slowing down in spatially patterned dryland ecosystems. *Ecography*, e06139, 2022. [10.1111/ecog.06139](https://doi.org/10.1111/ecog.06139)

- 24 **R. Martínez-García***, C.E. Tarnita, J.A. Bonachela. *Spatial patterns in ecological systems: from microbial colonies to ecosystem landscapes*. Biological (A)Symmetry Across the Scales on Emerging Topics in Life Sciences; ETL20210282 (2022). [D10.1042/ETLS20210282](https://doi.org/10.1042/ETLS20210282)
- 23 J. Nauta, P. Simoens, Y. Khaluf, **R. Martínez-García***. *Foraging behavior and patch size distribution jointly determine population dynamics in fragmented landscapes*. Journal of the Royal Society Interface 19 (191), 20220103, 2022. [10.1098/rsif.2022.0103](https://doi.org/10.1098/rsif.2022.0103)
- 22 **G.A. Maciel, R. Martínez-García***. *Enhanced species coexistence in Lotka-Volterra competition models due to nonlocal interactions*. Journal of Theoretical Biology, 530, 110872, (2021) [10.1016/j.jtbi.2021.110872](https://doi.org/10.1016/j.jtbi.2021.110872)
- 21 M.J. Noonan*, **R. Martínez-García**, G.H. Davis, M.C. Crofoot, R. Kays, B.T. Hirsch, E. Payne, A. Sih, D.L. Sinn, O. Spiegel, W.F. Fagan, C.H. Fleming, J.M. Calabrese. *Estimating encounter location distributions from animal tracking data*, Methods in Ecology and Evolution. 12(7), 1158-1173 (2021). [10.1111/2041-210X.13597](https://doi.org/10.1111/2041-210X.13597)
- 20 **R. Martínez-García***, C. López and F. Vazquez. *Species exclusion and coexistence in a noisy voter model with a competition-colonization tradeoff*. Physical Review E 103, 032406 (2021). [10.1103/PhysRevE.103.032406](https://doi.org/10.1103/PhysRevE.103.032406)
- 19 C. Cabal*, **R. Martínez-García**, A. de Castro, F. Valladares, S.W. Pacala, *Future paths for the exploitative segregation of plant roots*. Plant Signaling and Behavior, 1891755 (2021). [10.1080/15592324.2021.1891755](https://doi.org/10.1080/15592324.2021.1891755)
- 18 C. Cabal*, **R. Martínez-García**, A. de Castro, F. Valladares, S.W. Pacala, *The exploitative segregation of plant roots*, Science 370 (6521), 1197-1199 (2020). [10.1126/science.aba9877](https://doi.org/10.1126/science.aba9877)
- Journal cover and highlighted by national and international media (New York Times, Folha de São Paulo, El Mundo...)
- 17 **R. Martínez-García**, C.H. Fleming, R. Seppelt, W.F. Fagan and J.M. Calabrese*, *How range-residency and long-range perception change encounter rates*. J Theor Biol, 498, 110267, (2020). [10.1016/j.jtbi.2020.110267](https://doi.org/10.1016/j.jtbi.2020.110267)
- 16 F. W. Rossine*, **R. Martínez-García***, A.E. Sgro, T. Gregor*, C.E. Tarnita*. *Eco-evolutionary consequences of loners*. PLoS Biology, 18(3), e3000642 (2020). [10.1371/journal.pbio.3000642](https://doi.org/10.1371/journal.pbio.3000642)
- 15 E.H. Colombo*, **R. Martínez-García**, C. López, E. Hernández-García, *Spatial eco-evolutionary feedbacks mediate coexistence in prey-predator systems*. Scientific Reports, 9, 18161 (1-15) (2019). [10.1038/s41598-019-54510-6](https://doi.org/10.1038/s41598-019-54510-6)
- 14 **R. Martínez-García**, C.D. Nadell*, R. Hartmann, K. Drescher, J.A. Bonachela*. *Cell adhesion and fluid flow jointly initiate genotype spatial distribution in biofilms*. PLoS Comp Biol, 14(4): e1006094 (2018). [10.1371/journal.pcbi.1006094](https://doi.org/10.1371/journal.pcbi.1006094)
- 13 **R. Martínez-García** and C.E. Tarnita*. *Seasonality can induce coexistence of multiple bet-hedging strategies in Dictyostelium discoideum*. J Theor Biol, 426, 104-116 (2017). [10.1016/j.jtbi.2017.05.019](https://doi.org/10.1016/j.jtbi.2017.05.019)
- 12 **R. Martínez-García***, J.M. Calabrese and C. López. *Online games: a novel approach to explore how partial information influences human random searches*. Scientific Reports, 7, 40029 (2017). [10.1038/srep40029](https://doi.org/10.1038/srep40029)
- 11 **R. Martínez-García** and C.E. Tarnita*, *Lack of Ecological and Life History Context Can Create the Illusion of Social Interactions in Dictyostelium discoideum*. PLoS Comp Biol 12(12): e1005246 (2016). [10.1371/journal.pcbi.1005246](https://doi.org/10.1371/journal.pcbi.1005246)
- 10 **R. Martínez-García***, **C. Murgui**, E. Hernández-García and C. López. *Pattern formation in populations with density-dependent movement and two interaction scales*. PLoS ONE 10(7): e0132261 (2015). [10.1371/journal.pone.0132261](https://doi.org/10.1371/journal.pone.0132261)
- 9 **R. Martínez-García***, C. López and F. Vazquez. *Optimal recruitment strategies in groups of interacting walkers with leaders*. Phys Rev E, 91, 022117 (2015). [10.1103/PhysRevE.91.022117](https://doi.org/10.1103/PhysRevE.91.022117)
- 8 C.E. Tarnita*, A. Washburne*, **R. Martínez-García**, A.E. Sgro and S.A. Levin. *Fitness tradeoffs between spores and nonaggregating cells can explain the coexistence of diverse genotypes in cellular slime molds*. Proc. Natl. Acad. Sci. USA, 12(9), 2776-2781 (2015). [10.1073/pnas.1424242112](https://doi.org/10.1073/pnas.1424242112)
• Commentary by P.B. Rainey: *Precarious development: The uncertain social life of cellular slime molds*, Proc. Natl. Acad. Sci. USA, 112:2639-2640.

- 7 **R. Martinez-Garcia**[♦], J.M. Calabrese, E. Hernández-García, and C. López. *Minimal mechanisms for vegetation patterns in semiarid regions*. Philos. T. R. Soc. A, 372(2027): 20140068 (2014). [10.1098/rsta.2014.0068](https://doi.org/10.1098/rsta.2014.0068)
- 6 **R. Martinez-Garcia**[♦], J.M. Calabrese, and C. López. *Optimal search in interacting populations: Gaussian jumps versus Lévy flights*. Phys Rev E, 89, 032718, (2014). [10.1103/PhysRevE.89.032718](https://doi.org/10.1103/PhysRevE.89.032718)
- 5 **R. Martinez-Garcia**[♦], J.M. Calabrese, E. Hernández-García, and C. López. *Vegetation pattern formation in semiarid systems without facilitative mechanisms*. Geophys Res Lett, 40, 6143–6147, (2013). [10.1002/2013GL058797](https://doi.org/10.1002/2013GL058797)
- 4 **R. Martinez-Garcia**[♦], J.M. Calabrese and C. López. *Spatial patterns in mesic savannas: the local facilitation limit and the role of demographic stochasticity*. J Theor Biol, 333, 156-165, (2013). [10.1016/j.jtbi.2013.05.024](https://doi.org/10.1016/j.jtbi.2013.05.024)
- 3 **R. Martinez-Garcia**[♦], J.M. Calabrese, T. Mueller, K.A. Olson and C. López. *Optimizing the search for resources by sharing information: Mongolian gazelles as a case study*. Phys Rev Lett, 110, 248106, (2013). [10.1103/PhysRevLett.110.248106](https://doi.org/10.1103/PhysRevLett.110.248106)
 - Selected as Editor's suggestion and highlighted by the American Physical Society (Physics Focus), Physics World, the Spanish Royal Society of Physics and generalist press.
- 2 **R. Martinez-Garcia**[♦], F. Vázquez, C. López and M.A. Muñoz. *Temporal disorder in up-down symmetric systems*. Phys Rev E, 85, 051125, (2012). [10.1103/PhysRevE.85.051125](https://doi.org/10.1103/PhysRevE.85.051125)
- 1 S. Brouard[♦] and **R. Martinez**. *Cavity-mediated entanglement between distant atoms: Effect of spatial dispersion*. Phys Lett A. 375, 1640-1645, (2011). [10.1016/j.physleta.2011.03.004](https://doi.org/10.1016/j.physleta.2011.03.004)

Book chapters

- S. Vallina[♦], **R. Martinez-Garcia**, S.L. Smith, J.A. Bonachela. *Models in microbial ecology*. Encyclopedia of Microbiology, Elsevier. 4th edition (2019). [10.1016/B978-0-12-809633-8.20789-9](https://doi.org/10.1016/B978-0-12-809633-8.20789-9)
- C. López[♦], E. H. Colombo, E. Hernández-García, **R. Martinez-Garcia**. *Spatial competition and repulsion: pattern formation and the role of movement* in: "The Mathematics of Movement: an Interdisciplinary Approach to Mutual Challenges in Animal Ecology and Cell Biology" (Eds. Luca Giuggioli & Philip Maini) (in press).
- C.H. Fleming[♦], W.F. Fagan, J.M. Alston, G. Anand, N. Hiram, E.Y. Lin, **R. Martinez-Garcia**, M.J. Noonan, J.M. Calabrese. *Spatial Distributions for Animal Movement Processes* in: "The Mathematics of Movement: an Interdisciplinary Approach to Mutual Challenges in Animal Ecology and Cell Biology" (Eds. Luca Giuggioli & Philip Maini) (in press).

Preprints

- 7 J.O. Abraham^{*♦}, **R. Martinez-Garcia**^{*♦}, F. Gijssman[#], E.M. Phillips[#], C.E. Tarnita. *Should I stay or should I go? Modeling the decision-making process behind ungulate partial migration*. bioRxiv 2026.07.07.737075 [10.64898/2026.07.07.737075](https://doi.org/10.64898/2026.07.07.737075) (under review at Science Advances)
- 6 I. Braga, L. Wardil, **R. Martinez-Garcia**[♦], *Fixation probabilities for multi-allele Moran dynamics with weak selection*. arXiv 2604.12004 (2026) [10.48550/arXiv.2604.12004](https://doi.org/10.48550/arXiv.2604.12004) (revised and resubmitted to Phys Rev Research).
- 5 M. Tadej[♦], **R. Martinez-Garcia**, M. Hecht, *Extinction and persistence criteria in non-local Klausmeier model of vegetation dynamics on flat landscapes*. arXiv 2601.06681 [10.48550/arXiv.2601.06681](https://doi.org/10.48550/arXiv.2601.06681) (revised and resubmitted to Communications in Nonlinear Science and Numerical Simulation).
- 4 J. Liang[♦], O. Peleg, C. Hui, P. Reich, Y. Gu, **R. Martinez-Garcia**, *The architecture of forest intelligence: An evaluative framework*. (revised and resubmitted to PNAS Nexus).
- 3 A. Surendran, J.M. Calabrese, W.F. Fagan, & **R. Martinez-Garcia**[♦]. *Using transient encounter rates to quantify spatial patterns of home-range organization*. arXiv: 2510.13017 (2025). [10.48550/arXiv.2510.13017](https://doi.org/10.48550/arXiv.2510.13017)
- 2 B. Garcia de Figueiredo, J.M. Calabrese, W.F. Fagan, **R. Martinez-Garcia**[♦]. *Ergodicity shapes inference in biological reactions driven by a latent trajectory*. arXiv:2409.11433 (2024). [10.48550/arXiv.2409.11433](https://doi.org/10.48550/arXiv.2409.11433) (revised and resubmitted to Phys Rev Research)

- 1 M.J. Noonan[♦], **R. Martínez-García**, C.H. Fleming, B. Garcia de Figueiredo *et al.*, *The search behavior of terrestrial mammals*. bioRxiv 2022.12.31.521874 (2022). [10.1101/2022.12.31.521874](https://doi.org/10.1101/2022.12.31.521874) (revised and resubmitted to Nat. Comms.)

Teaching

Full semester courses.

For regular university courses, I specify the equivalence between the number of credits and hours taught, which varies across countries.

- 2025** *Course title:* Spatial models in environmental sciences (taught twice).
& *Program:* Environmental Modelling (M.Sc.),
2026 *Institution:* Carl von Ossietzky University, Oldenburg.
Level: Graduate. *Number of credits:* 3 (14 two-hour lectures).
- 2022** *Course title:* Stochastic Models in Biology and Ecology.
Program: Serrapilheira/ICTP-SAIFR training program in Quantitative Biology and Ecology.
Institution: Institute of Theoretical Physics, UNESP and Bioscience Institute, USP.
Level: Graduate. *Number of credits:* 6 (16 two-hour lectures).
- 2022** *Course title:* Advanced Topics in Biology and Ecology.
Program: Serrapilheira/ICTP-SAIFR training program in Quantitative Biology and Ecology.
Institution: Institute of Theoretical Physics, UNESP and Bioscience Institute, USP.
Level: Graduate. *Number of credits:* 12 (32 two-hour lectures).
- 2021** *Course title:* Deterministic and stochastic nonlinear dynamics: theory and applications.
Program: Master's in Theoretical Physics.
Institution: Instituto de Física Teórica / UNESP.
Level: Graduate. *Number of credits:* 12 (32 two-hour lectures).
- 2020** *Course title:* Non-linear dynamical systems and applications
Program: Master's in Theoretical Physics.
Institution: Instituto de Física Teórica / UNESP.
Level: Graduate. *Number of credits:* 6 (16 two-hour lectures)..
- 2019** *Course title:* Introduction to stochastic processes with applications in physics and biology.
Program: Master's in Theoretical Physics.
Institution: Instituto de Física Teórica / UNESP.
Level: Graduate. *Number of credits:* 6 (16 two-hour lectures)..
- 2015** *Course title:* EEB 325. Mathematical modelling in biology and medicine.
Program: Ecology and Evolutionary Biology.
Institution: Princeton University.
Level: Undergraduate. *Teaching load:* 10 hours, assisting Prof. Dr. Corina E. Tarnita.

Summer schools (invited lecturer)

- 2025** IFT-Perimeter-SAIFR Journeys into Theoretical Physics.
Course: Random walk theory in physics and biology.
Institution: ICTP-SAIFR, São Paulo, Brazil.
Level: Undergraduate. *Number of hours:* 5.
- 2023** Minicourse on Lattice models and applications to biological problems
Course: Spatial models in ecology.
Institution: ICTP-SAIFR, São Paulo, Brazil.
Level: Graduate and advanced undergraduate students. *Number of hours:* 6.
- 2022** II Summer School ICTP-SAIFR for Young Physicists.
Course: Introduction to statistical mechanics and its applications beyond physics.
Institution: ICTP-SAIFR, Institute for Theoretical Physics, UNESP, São Paulo, Brazil.
Level: High school. *Number of hours:* 4.5
- 2021** III School Jayme Tiomno in Theoretical Physics.
Course: Nonlinear phenomena in biology.
Institution: Institute of Physics, University of São Paulo, São Paulo, Brazil.
Level: Undergraduate. *Number of hours:* 10.

2019 IFT-Perimeter-SAIFR Journeys into Theoretical Physics.

Course: Nonlinear phenomena in biology.

Institution: ICTP-SAIFR, São Paulo, Brazil.

Level: Undergraduate.

Number of hours: 5.

• Mentoring & Curriculum development

Curriculum development.

2021- [ICTP – SAIFR / Instituto Serrapilheira Research training program in Quantitative Biology and Ecology.](#)

2022 Designed the program, including syllabus and faculty composition. Coordinated the first two editions, leading a four-person administration team. Annual budget: ~ 200.000 EUR.

Postdoc mentoring

2026 - Dr. Guilherme H. Costa

2025 - Dr. Rafael Menezes

2024 - Dr. David Pinto-Ramos

2023 - Dr. Anudeep Surendran.

2025-26 Dr. Ian Braga (now postdoctoral researcher at Academia Sinica, Taiwan).

2022-23 Dr. Pablo Souza de Castro (now Assistant Professor at the University of São Paulo, Brazil).

2021-22 Dr. Jesus M. Encinas Riveros (now postdoctoral researcher at Università di Padova, Italy).

2020-22 Dr. Vivian Dornelas (now Assistant Professor at Federal University of South of Bahia, Brazil).

2020-22 Dr. Gabriel Andreguetto Maciel (now data analyst at GTI It Solutions, Curitiba PR, Brazil).

PhD thesis supervised (url links direct to dissertation).

2026 *Student:* Mr. Paulo Ernesto de Cerqueira Junior

Title: A functional network approach to investigate dryland resilience.

Program: Physics. University of Göttingen.

Role: joint Prof. Dr. Viola Priesemann.

2025 *Student:* Mr. Zihao Liu

Title: Encounter statistics between cross-correlated stochastic processes.

Program: Physics. University of Göttingen.

Role: main supervisor.

2025 *Student:* Mr. Jelle van der Voort

Title: Resilience in complex systems through adaptive spatial pattern formation.

Program: Mathematics. University of Leiden.

Role: co-supervisor joint with Prof. Dr. Arjen Doelman and Prof. Dr. Max Rietkerk.

2025 *Student:* Ms. Melissa Fuentealba

Title: Mathematical models of fungus-plant interactions.

Program: PhD in Physics, University of Concepción (Chile)

Role: co-supervisor, joint with Dr. Ignacio Ormazábal.

2021-25 *Student:* Dr. Rafael Menezes (now postdoc in the group, located at ICTP-SAIFR, São Paulo).

Title: Not so well-mixed: Range-resident logistic model and movement-based crowding indexes [\[url\]](#).

Program: PhD in Ecology, Universidade de São Paulo, USP.

Role: main supervisor.

2022-24 *Student:* Dr. Nathan O. Silvano (now postdoc at IFISC, Spain).

Title: Non-equilibrium dynamics: phase transitions and pattern formation [\[url\]](#).

Program: PhD in Physics, State University of Rio de Janeiro, UERJ.

Role: co-supervisor.

Master thesis supervised (url links direct to dissertation).

2026- *Student:* Mr. Timon Althaus

Title: Competition-colonization tradeoffs and plant species coexistence in fragmented landscapes.

Program: Master's in Physics. Technical University of Dresden.

2025- *Student:* Mr. Khaled Yasein

Title: Quorum sensing and social dilemmas in microbial communities.

- Program:* Master's in Applied & Engineering Physics. Technical University of Munich.
- 2025-** *Student:* Mr. Vinicius Menuci.
Title: How seed.
Program: Master's in Theoretical Physics, IFT-UNESP.
- 2021-23** *Student:* Mr. João Pedro Valeriano (now PhD student at CENTURI - Université Aix Marseille).
Title: The effect of a chaotic environmental flow on the evolution of microbial social behaviors [\[url\]](#).
Program: Master's in Theoretical Physics, IFT-UNESP.
- 2021-23** *Student:* Mr. Daniel Cardoso Pereira Jorge (now PhD student at Princeton University).
Title: Aggregation patterns of living organisms: connecting mathematical modeling to data [\[url\]](#).
Program: Master's in Theoretical Physics, IFT-UNESP.
- 2020-23** *Student:* Mr. Benjamin Garcia (now PhD student at Princeton University).
Title: Absorption statistics of movement models with home-ranging behavior [\[url\]](#).
Program: Master's in Theoretical Physics, IFT-UNESP.
- 2019-20** *Student:* Mr. Jairo M. Rojas (now PhD student at University of Illinois Urbana-Champaign).
Title: Aging as a survival strategy in a microbial ecosystem [\[url\]](#).
Program: Master's in Theoretical Physics, IFT-UNESP.
- 2018** *Student:* Dr. Rodrigo Martínez (now postdoc at Donostia International Physics Center, DIPIC).
Title: A consumer-resource description of public-goods production in microbes [\[url\]](#).
Program: Master's in Physics of Complex Systems, IFISC.
- 2017** *Student:* Dr. Víctor Buendía (now senior postdoc at Bocconi University).
Title: Modeling quorum quorum-sensing mechanisms in bacterial populations [\[url\]](#).
Program: Master's in Physics of Complex Systems, IFISC.

Undergraduate students (not official theses).

- 2016** *Student:* Mr. David Pereñíguez (now postdoc at Bohr Institute, Univ of Copenhagen).
Project title: Aggregation patterns of living organisms: mathematical modeling & experiments.
Program: SURF summer internships, IFISC.
- 2013** *Student:* Dr. Clara Murgui (now tenure-track scientist at Institute of High-Energy Physics, Barcelona)
Project title: Aggregation in populations with density-dependent movement.
Program: IFISC-Intro summer internships, IFISC.

• Invited seminars and talks

Department and Institute colloquia (selected in the last three years).

- 2026** *Robustness and self-organization in Complex Living Systems.* Fudan University, Shanghai, China PR.
- 2026** *Robustness and self-organization in Complex Living Systems.* Centuri Seminar. Université Aix-Marseille. Marseille, France.
- 2026** *Robustness and self-organization in Complex Living Systems.* University of Rochester, Physics and Astronomy Colloquium. Rochester NY, USA.
- 2025** *From spatial patterns to functional resilience: dryland ecosystems as a case study.* Max Planck Institute for the Physics of Complex Systems, Institute Colloquium. Dresden, Germany.
- 2025** *Self-organization in drylands: from pairwise plant interactions to ecosystem robustness.* Max Planck Institute for Dynamics and Self-Organization, Institute Colloquium. Göttingen, Germany.
- 2025** *The range-resident logistic model: a new framework to understand how movement influences population dynamics.* Helmholtz Center for Environmental Sciences (UFZ). Leipzig, Germany.
- 2024** *Understanding belowground plant interactions mechanistically: from competition to facilitation.* Faculty of Physics and Earth System Sciences. University of Leipzig, Leipzig, Germany.
- 2024** *Ecological systems are rarely well-mixed: how movement behavior influences long-term ecological processes.* Institute for Chemistry and Biology of the Marine Environment. Carl von Ossietzky University, Oldenburg, Germany.
- 2024** *Lattice models applied to population dynamics: species coexistence and exclusion in a nonlinear noisy voter model.* Wroclaw University of Science and Technology. Wroclaw, Poland.

- 2024** *Causes and consequences of imperfect coordination in collective behaviors across scales: from microbial aggregates to ungulate migration.* Dutch Institute for Emergent Phenomena, Amsterdam, The Netherlands.
- 2024** *Causes and consequences of imperfect coordination in collective behaviors across scales: from microbial aggregates to ungulate migration.* University of Maryland, Mathematical Biology seminar. College Park, MD, USA.
- 2023** *Population dynamics models with non-local spatial interactions: from spatial patterns to species coexistence.* Mathematics colloquium, Swansea University, Swansea UK.
- 2023** *Ecological systems are rarely well-mixed.* Computational and Mathematical Biology seminar. University of Bath, England, UK.

Conference invited talks (selected in the last three years)

- 2026** *Vegetation patterns in heterogeneous drylands and their consequences for ecosystem resilience.* Mini-symposium on pattern formation in socio-ecological systems. Conference on Mathematical Models in Ecology and Evolution. Cork, Republic of Ireland.
- 2026** *Crowding indexes and population dynamics of range-resident animals: what they might say about seed dispersal.* Helmholtz Center for Environmental Research (UFZ). ERC Workshop. The dynamics of species-rich forests. Leipzig, Germany.
- 2026** *The role of transients in ecological regime shifts: dryland vegetation as a case study.* Workshop on transient dynamics: theory and applications. Potsdam Institute for Climate Change Research. Potsdam, Germany.
- 2025** *Nonreciprocal interactions induce early desertification in patterned ecosystems.* 16th Dynamical Systems applied to Biology and Natural Sciences conference. Naples, Italy.
- 2024** *Mathematical modeling of below-ground plant interactions: from competition to facilitation.* Dynamics Days Europe. Bremen, Germany.
- 2023** *Ecological dynamics with long-range individual interactions: from stochastic individual-based models to non-local partial differential equations.* XIX Workshop on Instabilities and Nonequilibrium Structures: Enrique Tirapegui 2023. Pontifical University of Valparaíso. Valparaíso, Chile.
- 2023** *Ecological dynamics with long-range individual interactions: from stochastic individual-based models to non-local partial differential equations.* ICMS workshop "Diffusive Systems: pattern formation, bifurcations, and biological applications". Edinburgh, UK.
- 2023** *Spatially explicit models of population dynamics: a dialogue between statistical physics and ecology.* Autumn Meeting of the Brazilian Physical Society. Federal University of Ouro Preto. Ouro Preto MG, Brazil.

- **Professional service:**

Editorial board

- 2025** – PLOS Ecosystems
- 2022** – PLOS Computational Biology
- 2017** – PLOS One

Society board member

- 2025** – Complex Systems Society (three-year tenure).

Journal reviewer (selected)

PNAS, Nature Communications, Science Advances, Physical Review Letters, Journal of Ecology, Ecology Letters, Methods in Ecology & Evolution, Ecography, Physical Review X, Physical Review X Life, eLife, ISME PLOS Computational Biology, Journal of the Royal Society Interface, Journal of Theoretical Biology, Physical Review Research, Physical Review E.

Funding agencies reviewer.

Instituto Serrapilheira, Brazil (2020); FAPESP, Brazil (2020 --); Human Frontiers Science Program (2022); Leverhulme Trust (2026).

Thesis committees.

External reviewer of 12 PhD theses and several Master's theses, both at my and external institutions worldwide. I have also served as a member of several (>15) qualification committees.

Event organization.

- 2026** School on Biological Physics across scales: phase transitions. ICTP-SAIFR, São Paulo, Brazil.
- 2024** School on Biological Physics across scales: pattern formation. ICTP-SAIFR, São Paulo, Brazil.
- 2024** Workshop on Applied Stochastic Processes for Encounter Problems. Brin Mathematics Research Center at the University of Maryland (approved). College Park MD, USA.
- 2023** Minicourse on Lattice models and applications to biological problems. ICTP-SAIFR, Sao Paulo, Brazil.
- 2022** ICTP Huddle: Eco-evolutionary Dynamics of Microbial Communities Across Scales. Trieste, Italy.
- 2021** [ICTP-SAIFR Physics Discussions](#). Series of online colloquia delivered by world-class researchers in all areas of physics (including Nobel Prize awardee Prof. Donna Strickland).
- 2020** Summer School in “Community Ecology: from patterns to principles”. ICTP-SAIFR, Sao Paulo, Brazil.

Participation in international cooperation events.

- 2026** 15th Sino-German Frontiers of Science and Technology Symposium (SINOGFOST) 2026: Advancing Resilience in Science, Health, and Technology. Suzhou (China). Only 60 participants (30 from Germany and 30 from China) were invited by the Alexander von Humboldt Foundation and the Chinese Academy of Sciences.

- **Outreach activities**

Mini-courses

- 2020,21** *Thermodynamics and introduction to statistical mechanics (translation from original title in Portuguese)* ICTP-SAIFR, São Paulo, Brazil.
20-hour course for high-school students.

Presentations (past five years)

- 2024** Patterns in Nature: a pretty coincidence or a search for survival? [original title in Portuguese: Padrões na natureza: uma bela coincidência ou uma procura pela sobrevivência?]. ICTP-SAIFR outreach talk.
- 2021** Biological complex systems: a physics approach to living systems. [original title in Portuguese: Sistemas biológicos complexos: um olhar da física para sistemas vivos]. Seminars of the Brazilian Academy of Sciences (talk in Portuguese).
- 2021** Mathematics and Ecology: do they need each other? [original title in Spanish: Matemáticas y ecología: ¿una relación necesaria?]. Online event organized by the Instituto Cervantes at Brasília (Brazil).
- 2020** *Forum Renascimento e Festival de Arte da Serrinha*. Round table about the 17 UN goals for sustainable development (Agenda 2030). (talk in Portuguese).
- 2020** *Mesa Ciência e Arte. Festival de Arte da Serrinha*. Round table with the participation of artists and scientists to discuss the connections between both disciplines. (talk in Portuguese).
- 2020** Complex Systems: when the whole is more than the sum of the parts. [original title in Portuguese: *Sistemas Complexos: quando o todo é maior do que a soma das partes.*] Física em casa. Series of outreach webinars organized by ICTP-SAIFR.

Articles

- 2021** [Por que o Nobel de Física representa o futuro da ciência](#). R. Martinez-Garcia. Blog Ciência Fundamental. Folha de São Paulo (in Portuguese).
- 2013** [La comunicación entre individuos podría optimizar la búsqueda de recursos en algunas especies](#). R. Martínez-García and C. López. Física Hoy, RSEF (in Spanish).